Day 5: Unsupervised Learning

Naeemullah Khan
naeemullah.khan@kaust.edu.sa

جامعة الملك عبدالله
للعلوم والتقنية
King Abdullah University of
Science and Technology

KAUST Academy
King Abdullah University of Science and Technology

April 2, 2026

Learning Outcomes

By the end of this session, you will be able to:

- Differentiate between Supervised and Unsupervised Learning paradigms.
- Understand K-Means clustering, and how it could be applied to discover hidden groups in data.
- Analyze high-dimensional data using PCA and t-SNE for visualization and compression.
- Understand how Autoencoders utilize neural networks for tasks like denoising, generation, and anomaly detection.

Table of Contents

1. Unsupervised Learning
2. Clustering: K-Means
3. Dimensionality Reduction: PCA
4. Dimensionality Reduction: t-SNE
5. Autoencoders
6. Industry Use Cases
7. References

What is Unsupervised Learning?

So far: We've been doing **Supervised Learning**, learning from labeled data (x, y).

Supervised Learning

Data: (x, y) pairs
- x: input (features)
- y: label (target)

Goal: Learn $f(x) \approx y$

Examples:
- Image classification
- House price prediction

Unsupervised Learning

Data: Only x (no labels!)
- Just raw data
- No target to predict

Goal: Find hidden patterns or structure

Examples:
- Customer segmentation
- Data compression

Key difference: No labels $\rightarrow$ algorithm must discover structure on its own!

Why Unsupervised Learning?

Labels are expensive!

- Labeling data requires human effort, time, and expertise
- Most real-world data is unlabeled
- Sometimes we don't even know what labels to look for

The Big Idea

Can we learn useful things from data **without** labels?
Yes! Unsupervised learning finds hidden patterns, structure, and representations.

Common Unsupervised Learning Tasks

What can we do without labels?

1. Clustering
Group similar data points together
Example: Customer segmentation, document group

2. Dimensionality Reduction
Compress high-dimensional data to lower dimensions
Example: Visualization, feature extraction

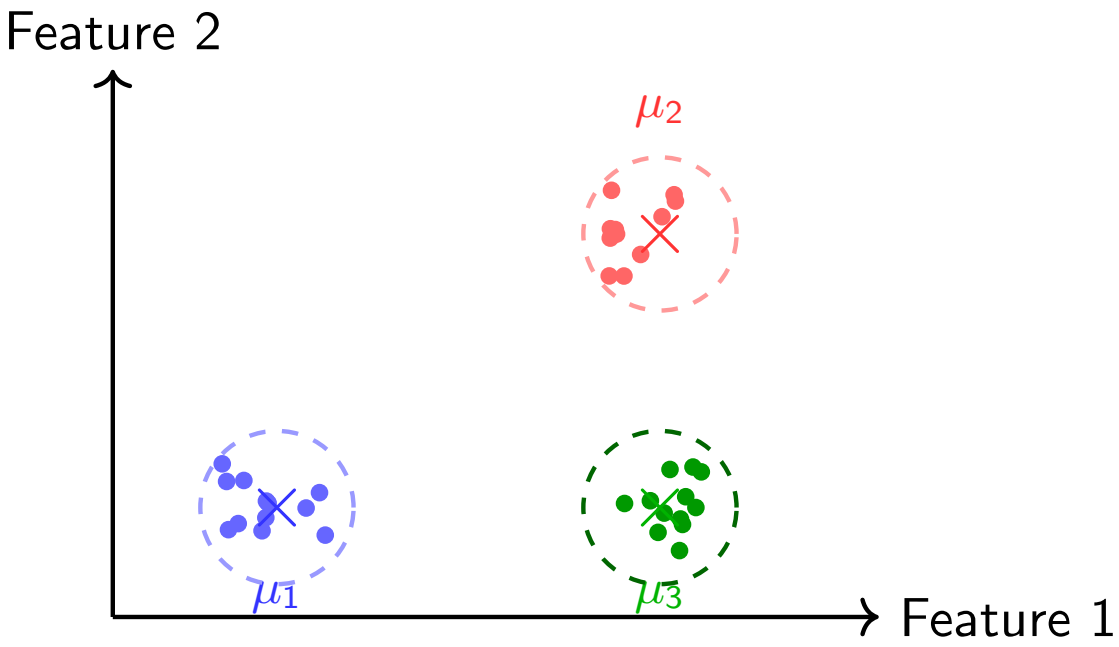
3. Anomaly Detection
Identify unusual or outlier data points
Example: Fraud detection, quality control

4. Generation
Learn to create new, realistic data samples
Example: Image synthesis, data augmentation

We'll explore four powerful algorithms that tackle these tasks.

Clustering: K-Means

Main Idea: Represent each cluster by its center (mean)



- Each cluster has a **center** (centroid) μ_k (marked with $\times$)
- Points belong to their **nearest** center
- Goal: Find centers that minimize total distance to assigned points

K-Means Algorithm

K-Means Algorithm

Input: Data $\{x_1, x_2, \ldots, x_n\}$ and number of clusters K

Initialize: Randomly place K cluster centers $\mu_1, \mu_2, \ldots, \mu_K$

Repeat until convergence:

1. **Assignment Step:** For each point x_i, assign to nearest center

$$c_i = \arg\min_{k=1,\ldots,K} \|x_i - \mu_k\|^2$$

(find which center is closest)

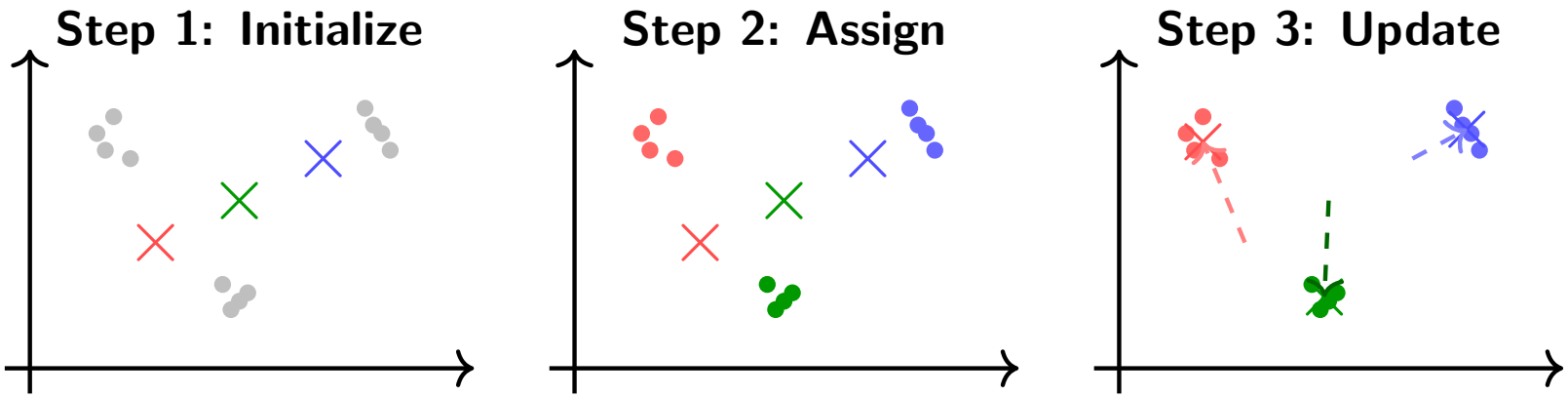
2. **Update Step:** Move each center to mean of assigned points

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i$$

(average all points in cluster k)

Stop when: Assignments don't change OR max iterations reached

K-Means: Step-by-Step Example



Repeat steps 2-3 until centers stop moving (typically 10-100 iterations)

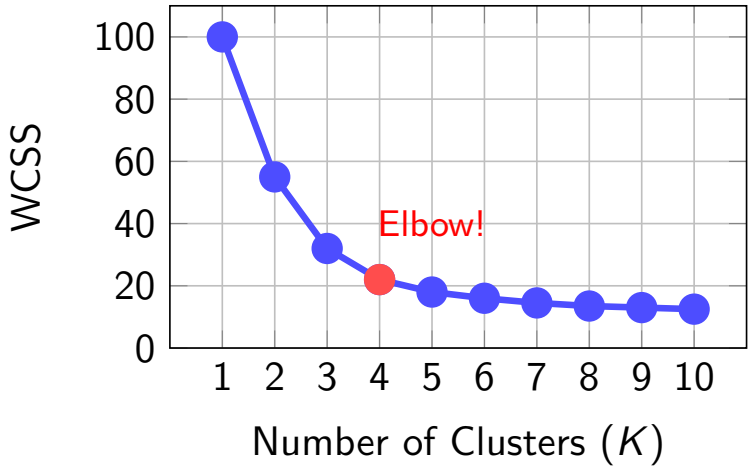
How to Choose K?

Problem: How many clusters should we use?

Elbow Method

Plot "within-cluster sum of squares" (WCSS) vs K

$$\text{WCSS} = \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2$$



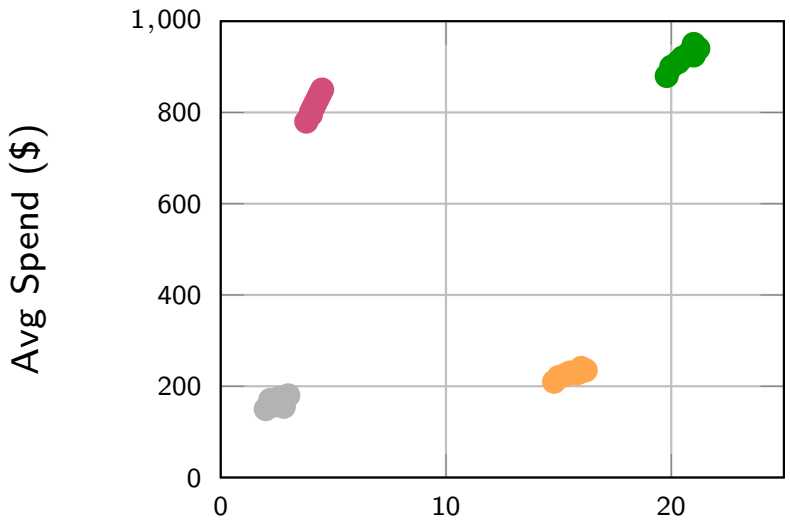
Look for the "elbow" where adding more clusters doesn't help much (in the above case it is 4)

Application 1: Customer Segmentation

Use case: Group customers for targeted marketing

Example: E-commerce Data

- Features: frequency, spend, recency
- Run K-Means with $K = 4$
- Interpret clusters



Discovered Segments:

Inactive
Low freq, low spend
Action: Re-engage

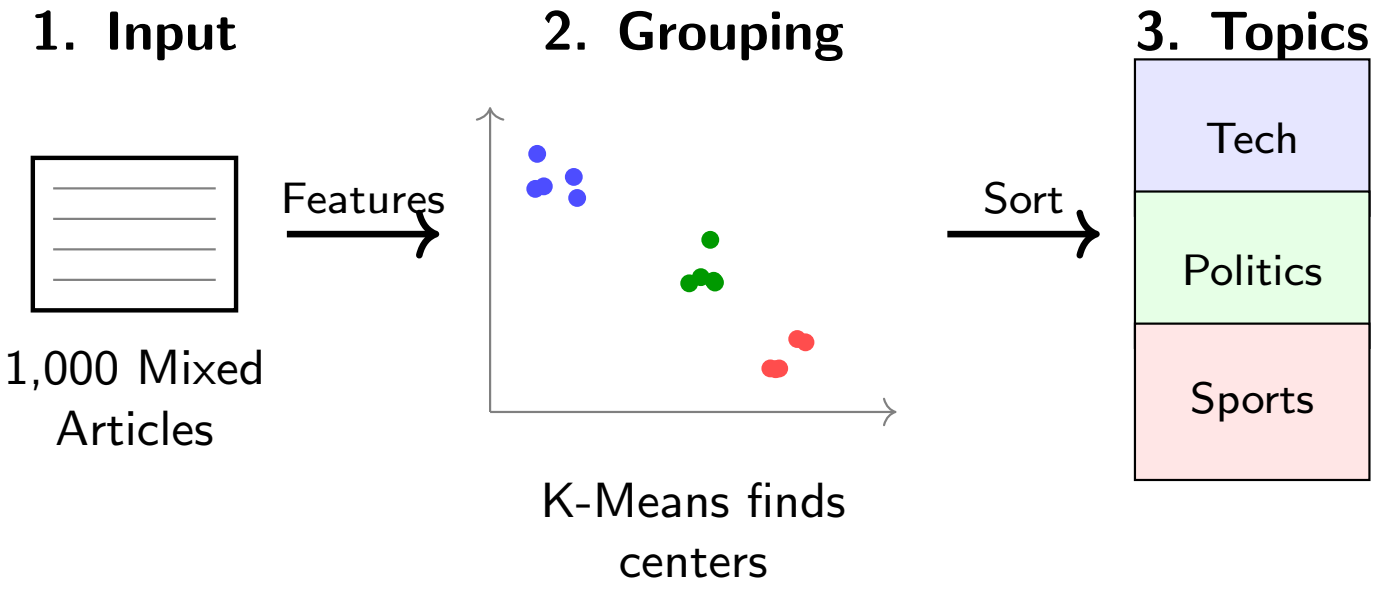
Bargain
High freq, low spend
Action: Discounts

Big Spenders
Low freq, high spend
Action: Premium Service

VIP
High freq, high spend
Action: Loyalty Program

Application 2: Document Clustering

Use case: Automatically organize news articles by topic.



Process

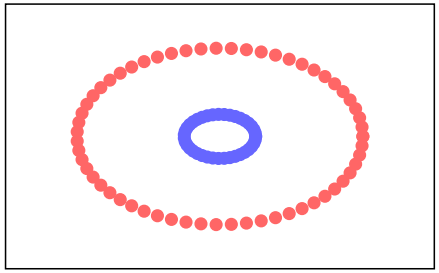
- Convert text into numbers (Text Features).
- Similar stories appear close together in space.
- K-Means automatically groups them into topics.

Example: Google News grouping stories from different sources.

When K-Means Fails

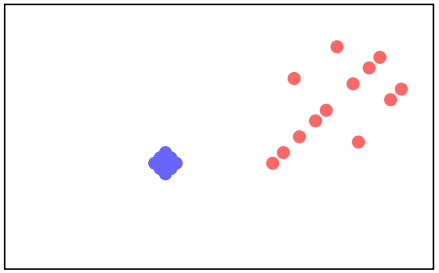
K-Means assumes spherical, equal-sized clusters

Problem 1: Non-spherical



Fails on concentric circles!

Problem 2: Different sizes



Struggles with size differences!

Other Limitations

- Sensitive to outliers
- Must specify K in advance
- Random initialization matters
- Bad results for high dimensional data

Alternatives: DBSCAN (density-based), Gaussian Mixture Models, Hierarchical clustering

The Curse of Dimensionality

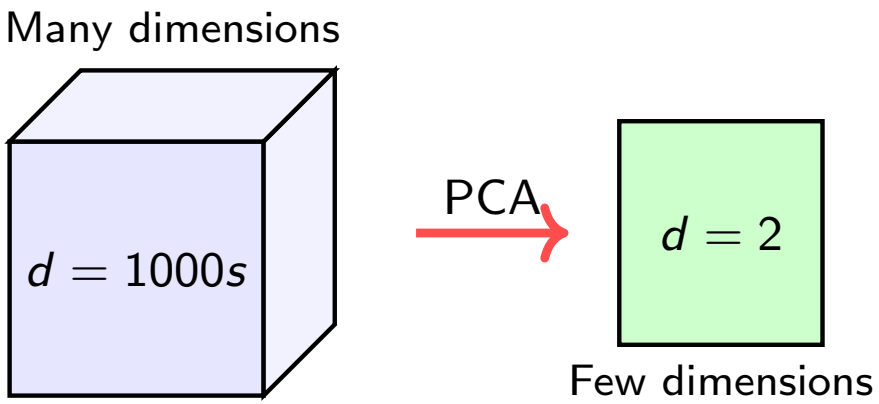
Real-world data is high-dimensional:

Examples:

- Images: $1000 \times 1000 = 1\text{M}$ pixels
- Text: 10,000+ words
- Genomics: 20,000+ genes
- Sensors: 100+ measurements

Problems:

- Hard to visualize
- Slow to compute
- Lots of redundancy
- Noisy measurements



Solution: Find a lower-dimensional representation while keeping important information (Dimensionality Reduction)

What is PCA?

Principal Component Analysis (PCA) is a dimensionality reduction algorithm that finds the most important directions in your data.

Core Idea

Instead of using original features, create **new features** (principal components) that:

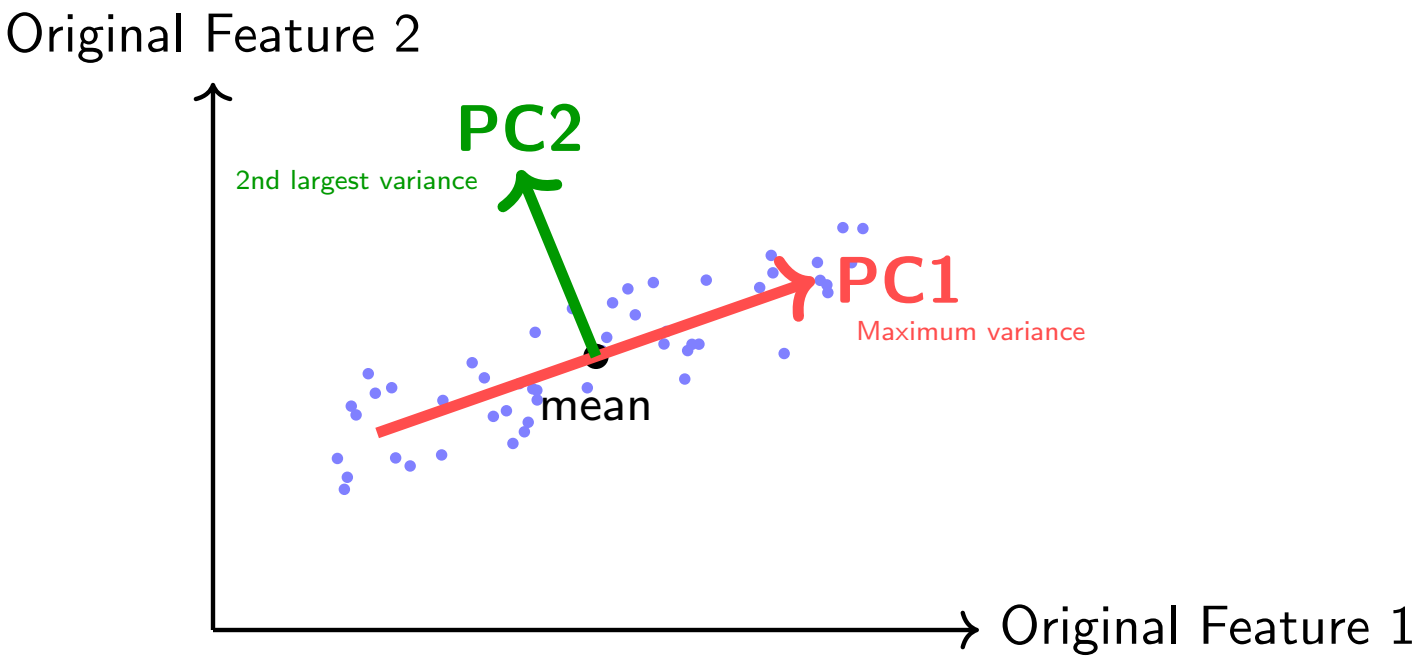
- Capture maximum variance
- Are uncorrelated with each other
- Are ranked by importance

Then keep only the top few components!

Original Data 784 features (correlated, noisy)	$\xrightarrow{\text{PCA}}$	New Data 50 components (uncorrelated, clean)

PCA Intuition: Finding the Best Directions

Goal: Find directions where data varies the most.



- PC1: Direction with most spread
- PC2: Perpendicular to PC1, next most spread
- PC3, PC4, ... : Each perpendicular to all previous ones

How Does PCA Work? (Part 1)

Step 1: Prepare the Data

1. Center the Data

Shift the data so the center is at the origin (0,0).

$$X_{\text{centered}} = X - \bar{X}$$

(Subtract the mean from every feature)

2. Compute Covariance Matrix

Calculate how features vary with respect to each other.

$$C = \frac{1}{n} X_{\text{centered}}^{\top} X_{\text{centered}}$$

(Finds correlations between features)

How Does PCA Work? (Part 2)

Step 2: Find Directions & Project

3. Find Principal Components

Calculate **Eigenvectors** and **Eigenvalues** of the covariance matrix C.

- **Eigenvectors:** The directions of the new axes.
- **Eigenvalues:** The amount of variance (information) along those axes.

4. Select & Project

Pick the top k eigenvectors with the largest eigenvalues and transform the data.

$$Z = X_{\text{centered}} \cdot V_k$$

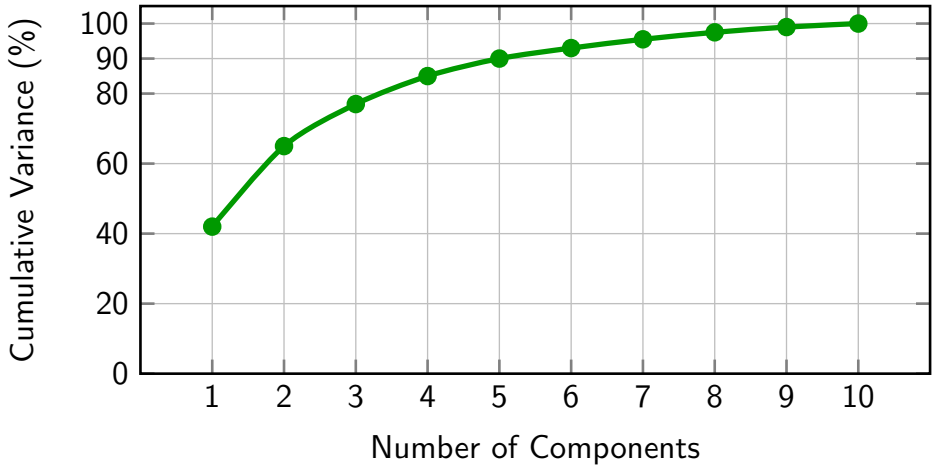
(Reduces data from d dimensions $\rightarrow$ k dimensions)

How Many Components to Keep?

Key question: How many components are enough?

Cumulative Variance Explained

(The most common method for selecting K)



Rule of Thumb: Choose the smallest K that preserves **80–95%** of the total variance.

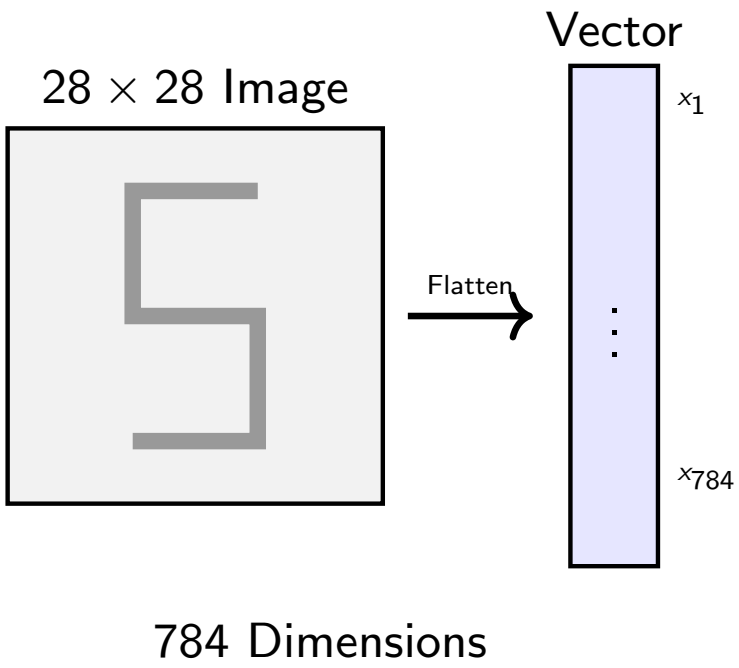
Application 1: Data Visualization

The Data: MNIST Digits, 70,000 handwritten digits (0–9).

The Challenge:

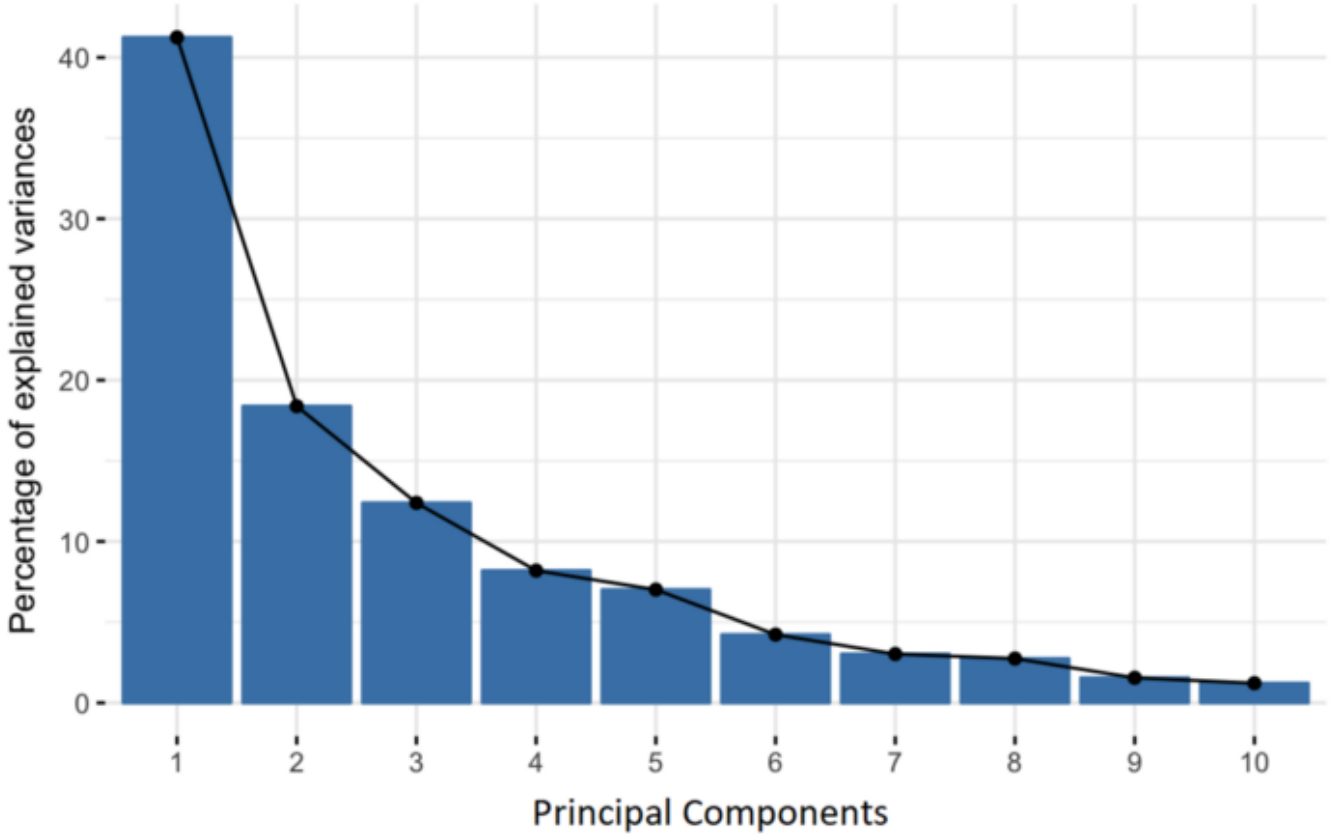
- Each image is 28×28 pixels.
- Total features: $28 \times 28 = \mathbf{784}$ dimensions!
- We cannot "see" in 784 dimensions.

Question: Can we compress this to just **2D** to see how the digits relate?



Application 1: Data Visualization

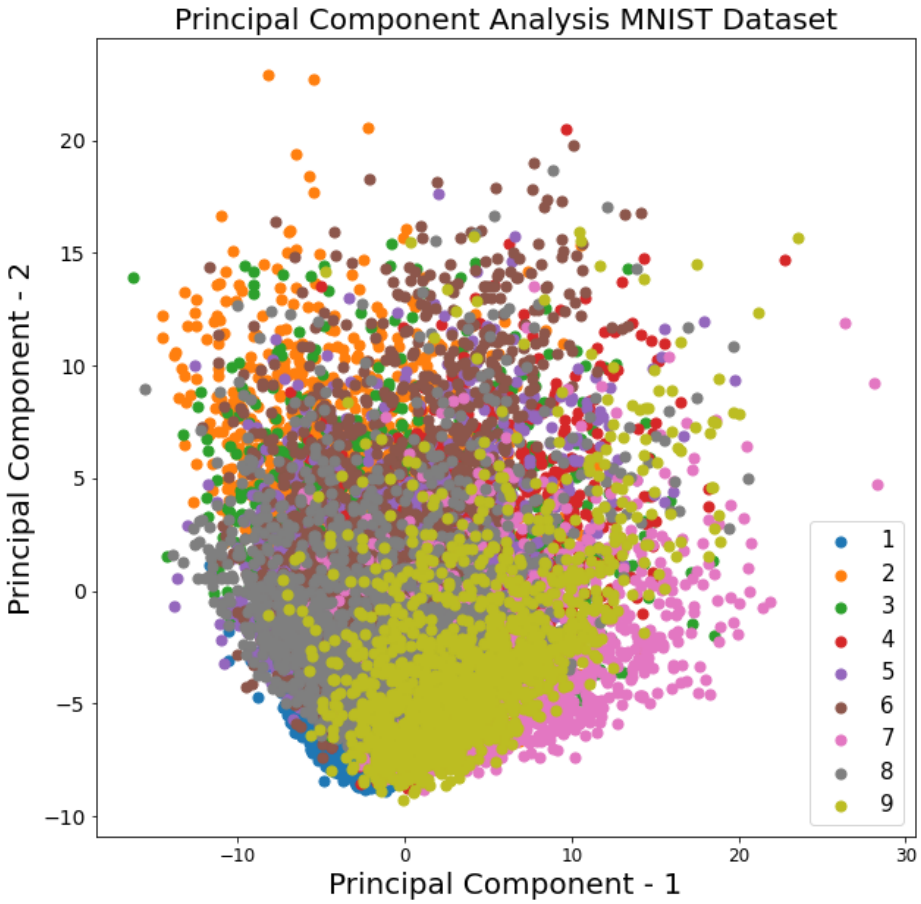
Applying PCA: We project the 784 dimensions down to the top 2 Principal Components.



Application 1: Data Visualization

Is this the best visualization?

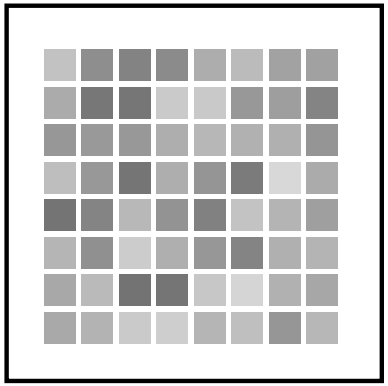
- It captures some structure, but points overlap heavily.
- **Reason:** PCA is **linear**, it cannot unfold complex, curved structures.
- PCA is mostly used for compression. We will cover a better visualization algorithm shortly.



Application 2: Data Compression

Use case: Store data more efficiently

Original



$28 \times 28 = 784$ values

PCA
50 comps

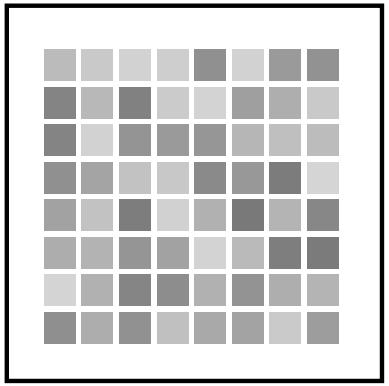
Compressed

[0.42, -1.3, 0.87, ...]
50 values

93% smaller

Reconstruct

Reconstructed



Similar to original
(but with small error)

More components = better reconstruction, less compression

10 PCs	**50 PCs**	**200 PCs**
99% compression	93% compression	75% compression
(blurry)	(good)	(near perfect)

PCA Limitations

PCA Limitations:

- **Linear only**: Finds linear combinations of features
- **Sensitive to scale**: Must standardize data first
- **Not Interpretable**: PCs are combinations of all features
- **Outliers**: Can distort principal components

Let's have a look at a better algorithm for high dimensional data visualization...

t-SNE: Why Do We Need It?

PCA is great... but it's linear. Many datasets have **non-linear** structure.

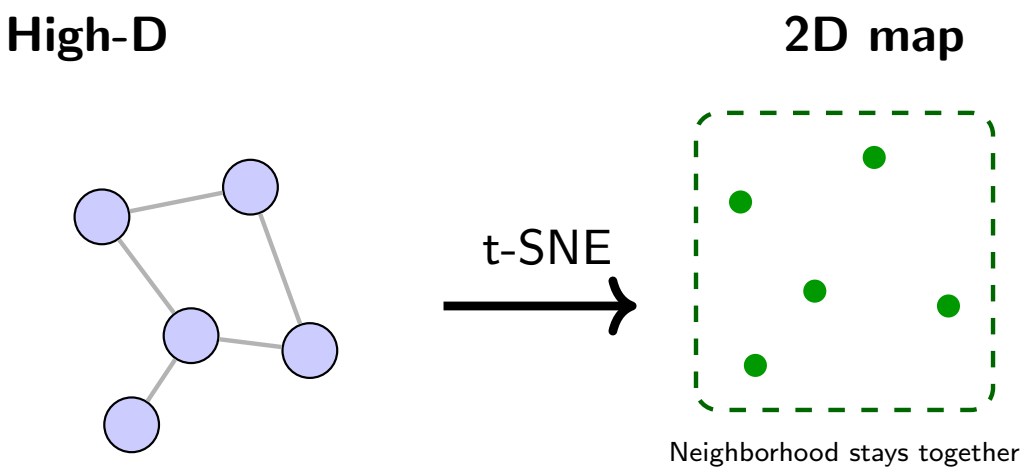
Problem

- High-dimensional data can be hard to visualize
- PCA may **mix** classes if the separation is non-linear
- We mainly want a **good 2D plot** for exploration

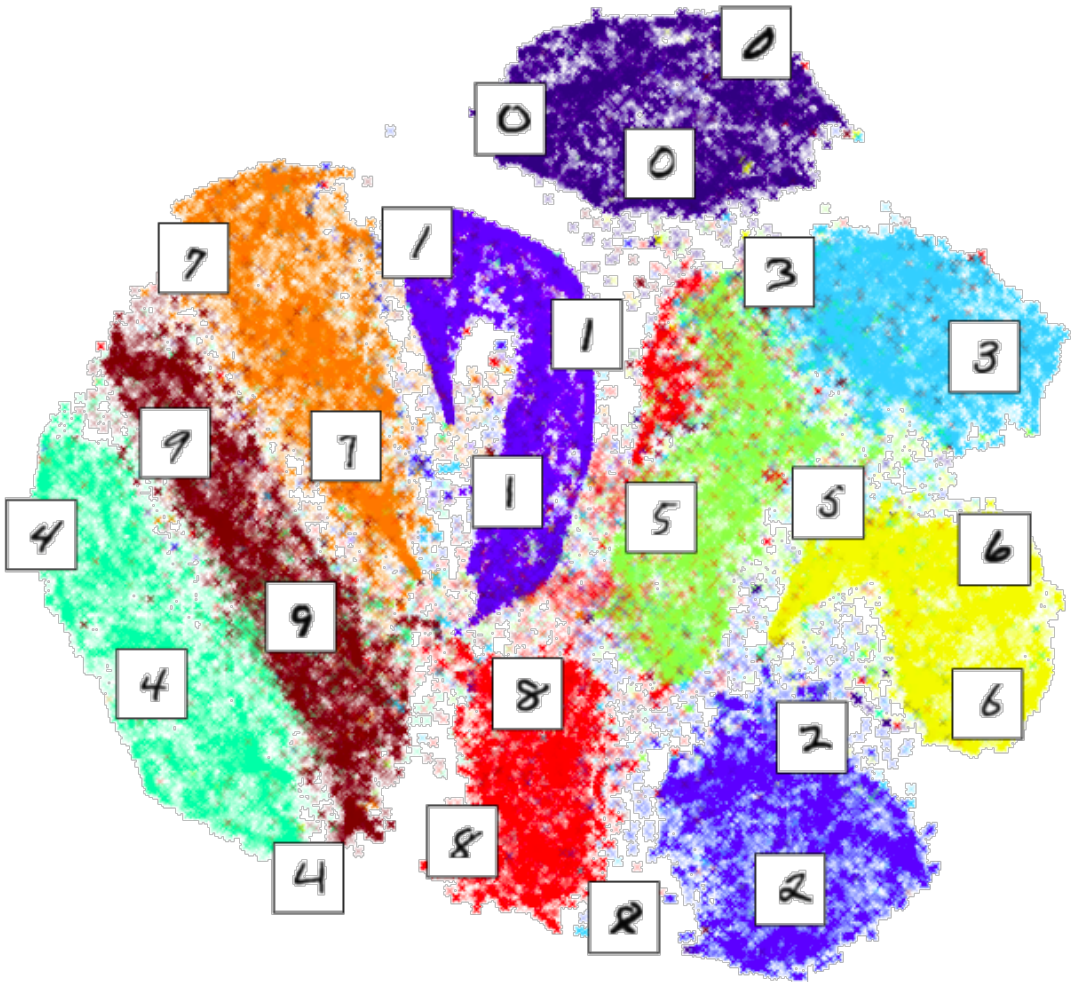
t-Distributed Stochastic Neighbor Embedding (t-SNE)

Main Idea

t-SNE tries to build a 2D map where: neighbors in high-D remain neighbors in 2D.



Example: Visualizing MNIST via t-SNE



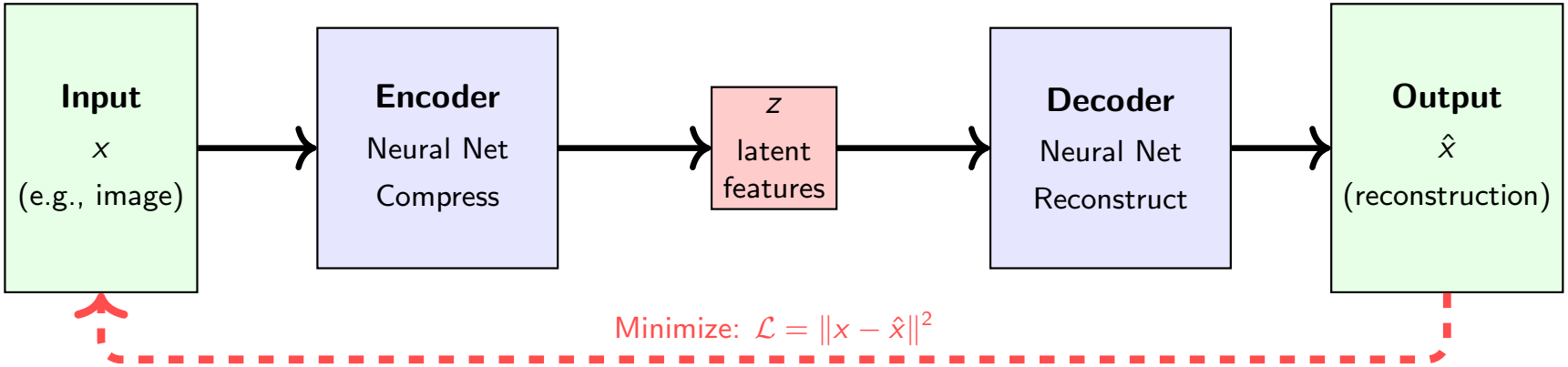
This is much better than PCA! We can see how similar digits are grouped together, while different ones are far from each other.

We have covered classical algorithms like K-Means and PCA.

Now, let's explore a powerful Neural Network architecture designed for unsupervised learning...

What is an Autoencoder?

Autoencoder: a type of neural network that learn to compress data into a compact form and then reconstruct it to match the original input.



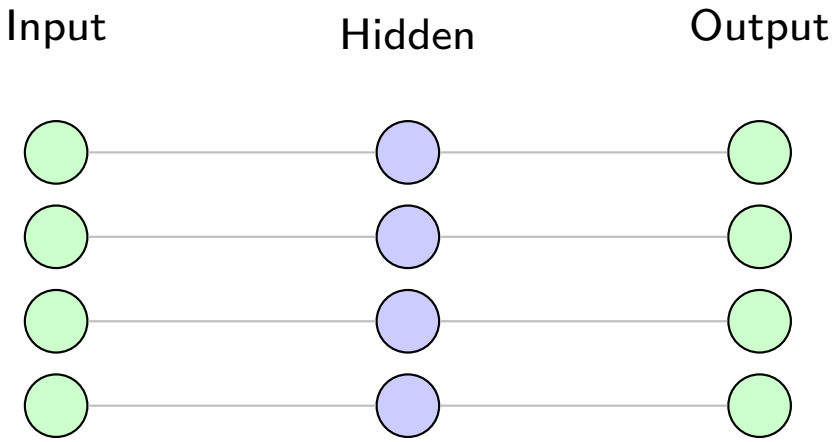
The trick: Force information through a narrow **bottleneck** z.

This forces the network to learn compressed, meaningful representations!

Why Does the Bottleneck Work?

What if there's no bottleneck?

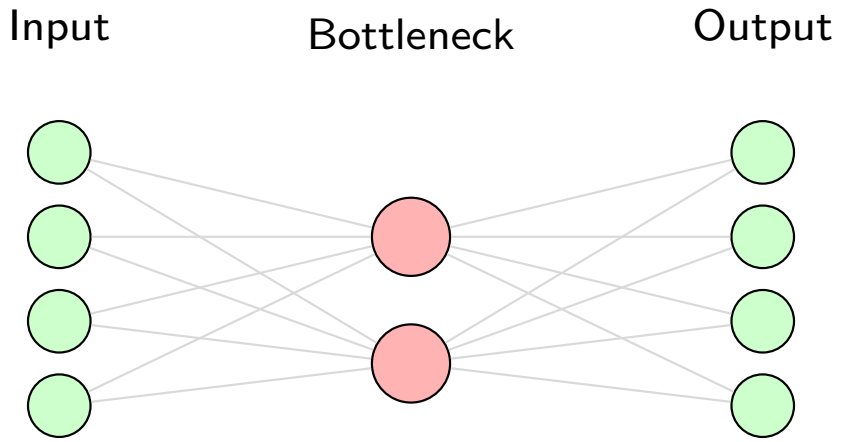
Without Bottleneck



Network can just copy:
$h_i = x_i$, $\hat{x}_i = h_i$

Learns nothing useful!

With Bottleneck

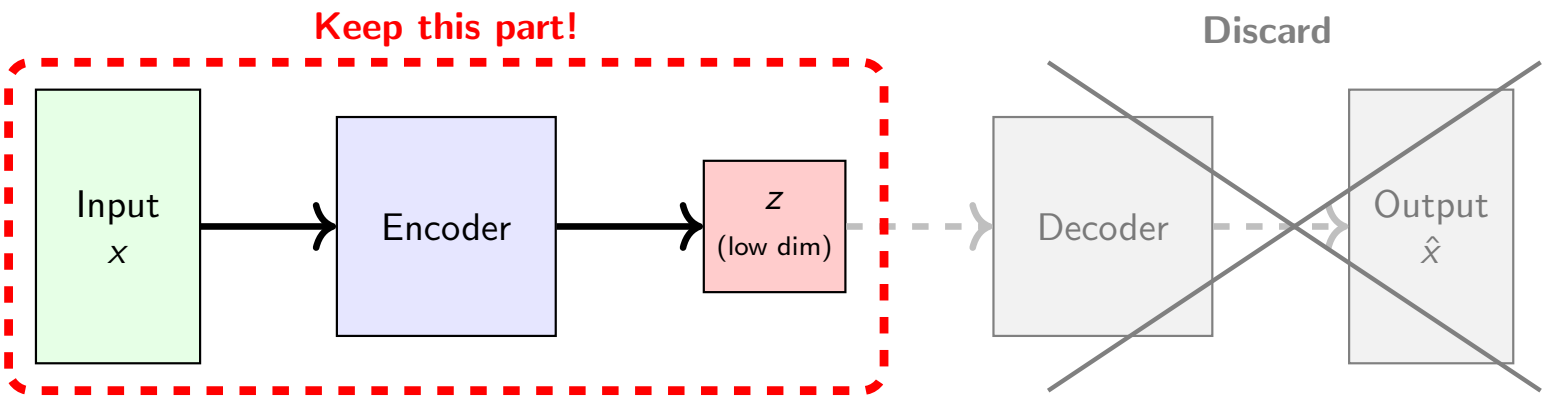


Must compress 4D $\rightarrow$ 2D
Forces learning of structure!

Learns meaningful features!

Application 1: Dimensionality Reduction

How to use an Autoencoder for compression:

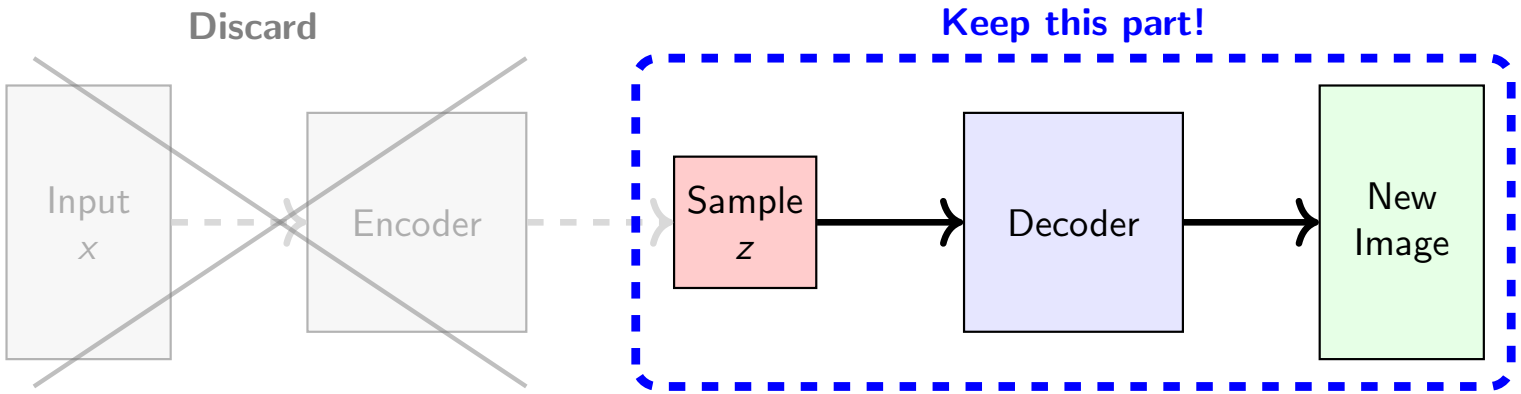


The Process

1. **Train** the full network to reconstruct the input ($x \approx \hat{x}$).
2. **Discard** the Decoder (it's only needed for training).
3. **Use the Encoder** to map high-dimensional data x to low-dimensional code z.

Application 2: Image Generation

How to use an Autoencoder for generation:

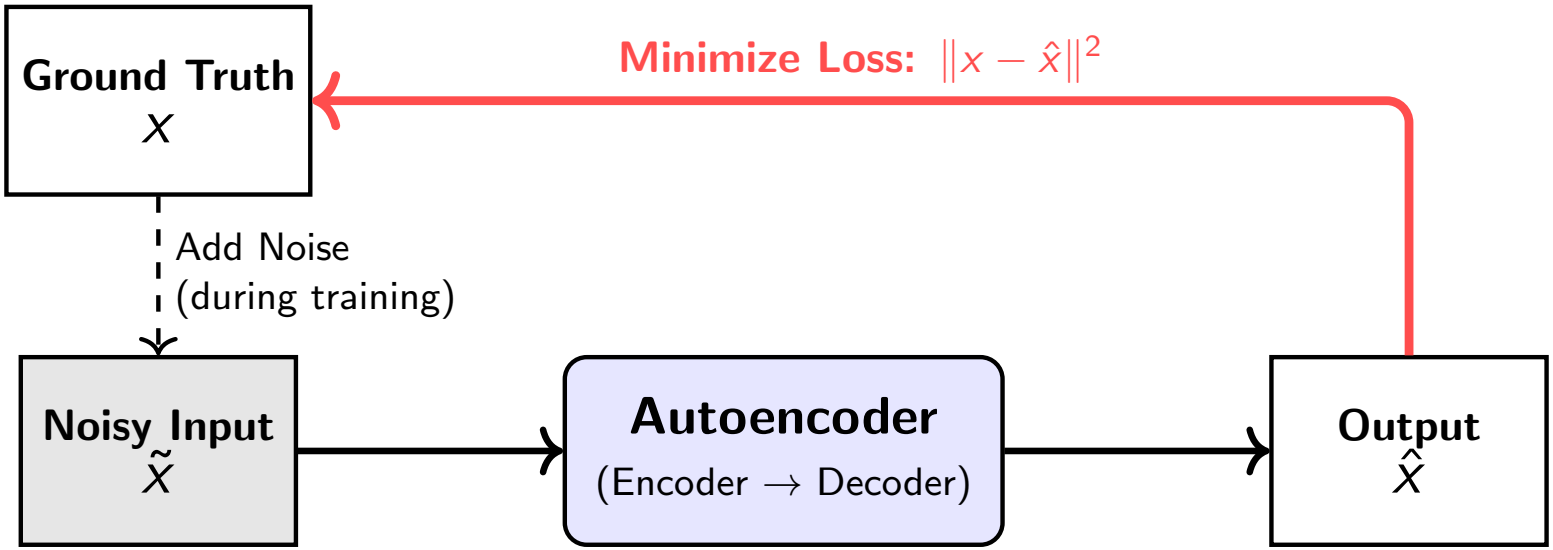


The Process

1. **Train** the full network on images to learn features.
2. **Discard** the Encoder.
3. **Sample** a random vector z (pick random numbers).
4. **Generate** a brand new image by running z through the Decoder.

Application 3: Denoising Autoencoders

Idea: Train the network to recover the original signal from corrupted input.



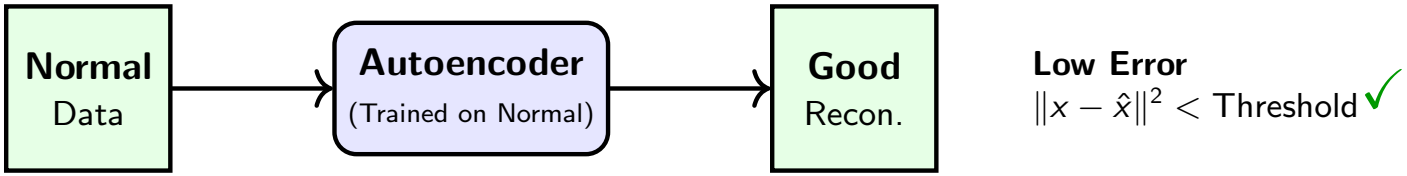
How it works

- Take clean image x.
- Corrupt it to get $\tilde{x}$.
- Force AE to map $\tilde{x} \rightarrow x$.

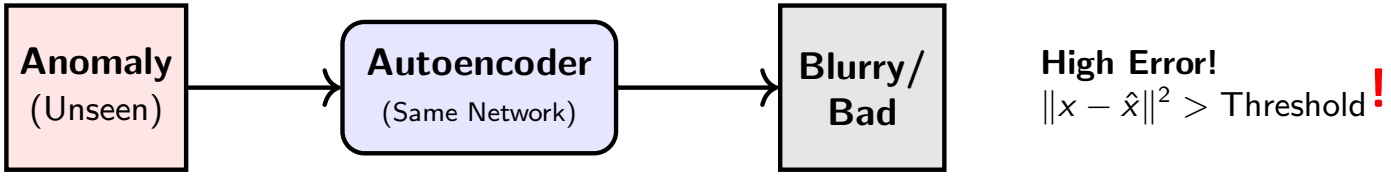
Application 4: Anomaly Detection

Concept: The network learns to reconstruct "Normal" data perfectly. When it sees an anomaly, it fails to reconstruct it.

1. Training / Normal Operation



2. Detection Phase



Real-World Use Cases

- **Credit Card Fraud:** Normal transactions reconstruct well, theft looks weird.
- **Manufacturing:** Detect defective parts on an assembly line.

Use Case 1: Equipment Health Monitoring

Problem: Unplanned failures cost $250K–500K per hour; abnormal events take days to diagnose.

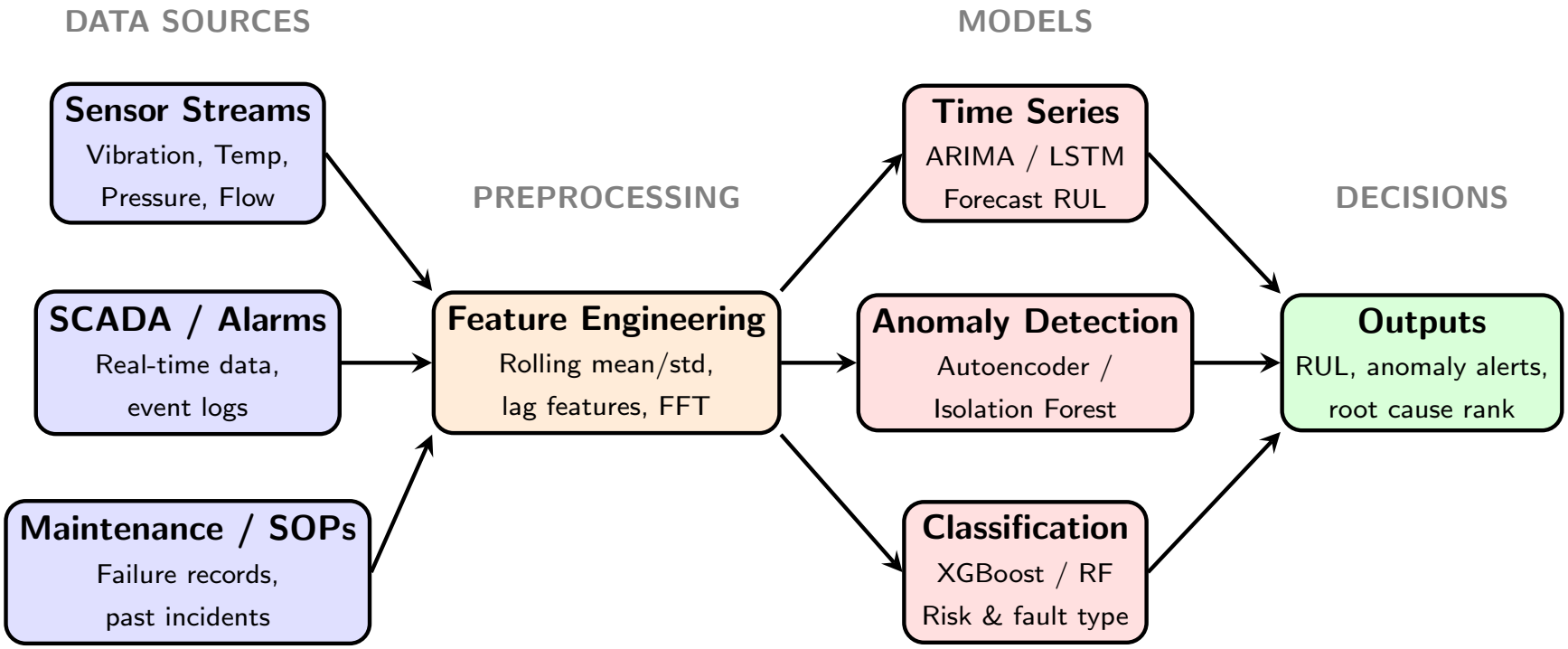
AI Solution: A unified monitoring system that **predicts** future failures and **diagnoses** current anomalies.

Shell: 20% less unplanned downtime across 10,000+ assets **BP:** $100M+ annual savings 70–80% faster root cause identification

Input Data:

- Sensor streams (vibration, temperature, pressure, flow, acoustic)
- SCADA / historian data, alarm & event logs, operator actions
- Maintenance records, failure logs, SOPs, past incident reports

Equipment Health Monitoring: AI Pipeline



Equipment Health: Prediction Side

"When will it fail?" — forward-looking models for maintenance planning:

Model	What it does	Why it matters
Time-Series ARIMA / LSTM	Forecasts future sensor trajectories, e.g. vibration trend over the next 4 weeks	Answers *"when will this reading cross the danger threshold?"*—gives a Remaining Useful Life (RUL)
Risk Classification XGBoost / Random Forest	Assigns a failure probability and failure-mode label to each asset based on current features	Prioritises *which* equipment to inspect first and *what* failure mode to look for

Equipment Health: Diagnosis Side

"What is happening and why?" — reactive models for real-time response:

Model	What it does	Why it matters
Anomaly Detection Autoencoder / Isolation Forest	Flags readings that deviate from the learned healthy operating pattern	Catches novel or compound failures that the classifier has never seen—no labels needed
Lag Analysis Cross-corr. / LSTM	Computes lagged correlations between variables to find which signal moved first	Answers *"did the temperature spike cause the pressure drop, or vice versa?"*
Incident Matching Similarity / Clustering	Matches the current event signature against a database of resolved past incidents	Retrieves proven fixes—engineers don't start from scratch each time

Use Case 2: Production Optimization

Problem: Manual control leaves 10–15% efficiency on the table; suboptimal conditions waste energy.

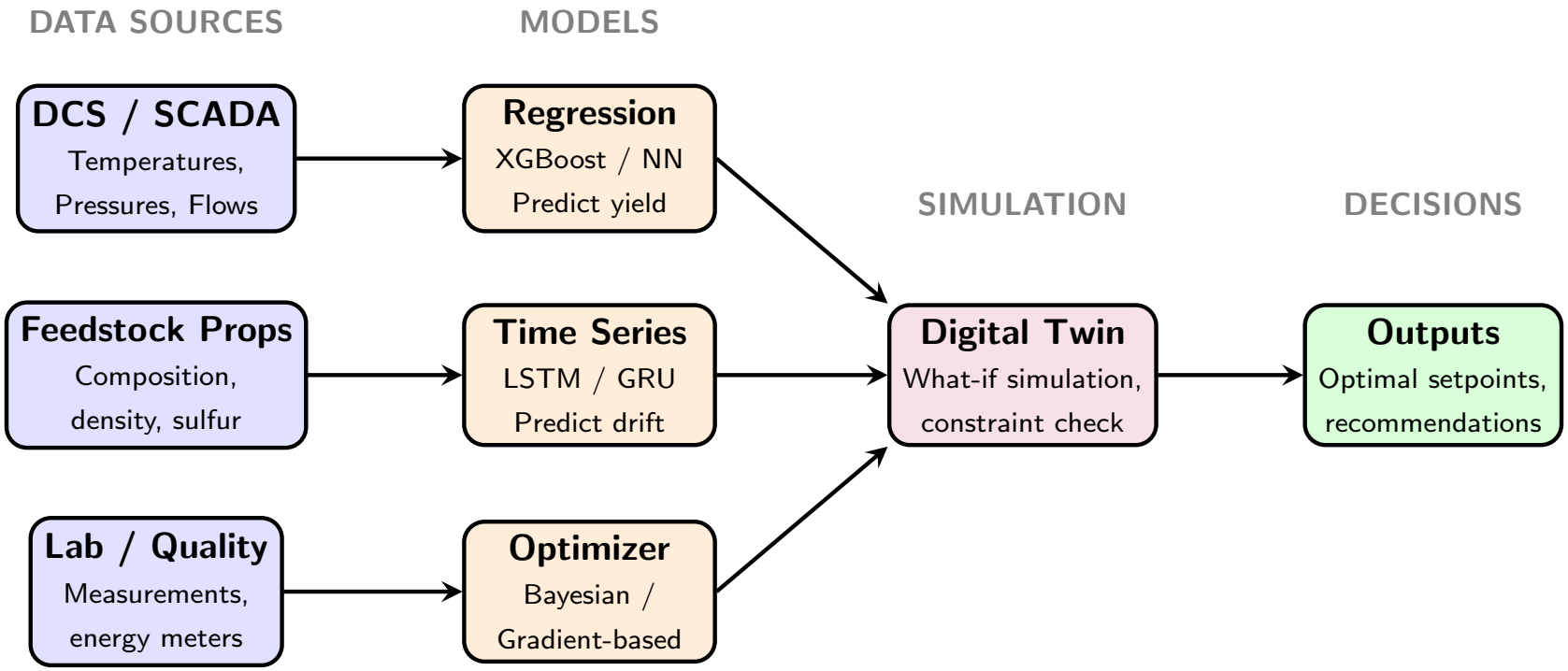
AI Solution: Real-time ML optimizes 100+ process variables simultaneously via digital twin simulation.

BP: 20% refining efficiency increase
Aramco Khurais: 15% production increase from same assets

Input Data:

- DCS/SCADA variables (temperatures, pressures, flows, compositions)
- Feedstock properties, equipment constraints
- Lab / quality measurements, energy consumption data

Production Optimization: AI Pipeline



Production Optimization: Model Layer

Three components work together to find optimal operating conditions:

Model	What it does	Why it matters
Regression XGBoost / Neural Net	Predicts yield, conversion, and energy usage from current process variables	Answers *"if I change temperature by 5°C, what happens to yield?"*
Time Series LSTM / GRU	Forecasts process drift and future plant state over the next hours	Catches slow degradation before it impacts product quality
Optimizer Bayesian / Gradient	Searches the setpoint space to maximise throughput within constraints	Finds the best operating point that humans would not try manually

Use Case 3: AI in Advanced Materials

Polymer / Silicon Products

Problem: New product development takes 3–5 years and costs \$10–50M; trial-and-error testing wastes materials and time.

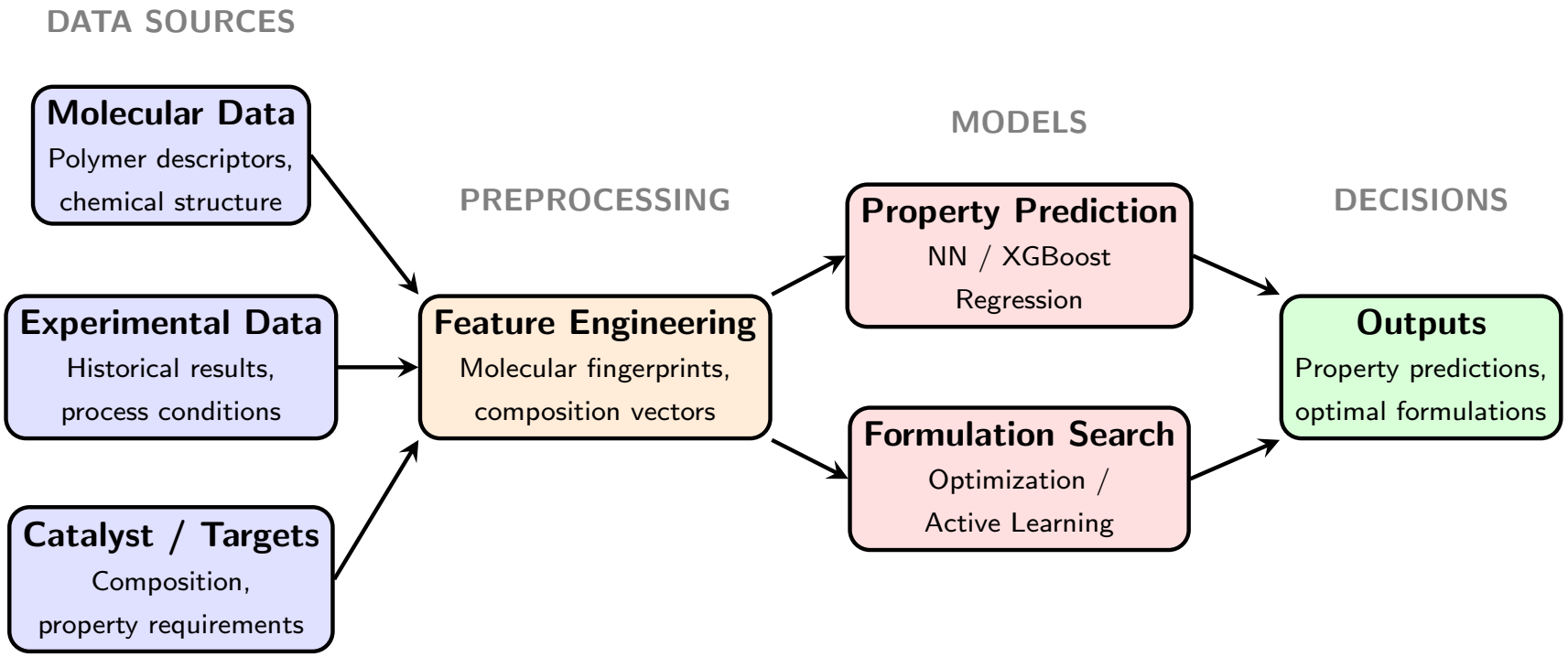
AI Solution: ML predicts material properties directly from molecular structure—screening 10× more candidates than traditional methods.

50–70% reduction in development time 10× more formulations screened Discovery of novel materials not found by human intuition

Input Data:

- Molecular / polymer descriptors, chemical structure
- Historical experimental data, process conditions
- Catalyst composition, target property requirements

AI in Advanced Materials: AI Pipeline



AI in Advanced Materials: Model Layer

Regression models replace years of lab trial-and-error:

Model	What it does	Why it matters
Property Prediction NN / XGBoost	Predicts strength, conductivity, and thermal stability from molecular descriptors and composition	Screens thousands of candidates in seconds instead of months of lab work
Formulation Search Bayesian / Active Learning	Searches the composition space to find formulations that meet target properties within constraints	Finds optimal catalyst and polymer compositions that human intuition would miss

References

- Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*
- Andrew Ng, *Machine Learning Specialization* (Coursera/DeepLearning.AI)

Slides contributed by Mohamed Eltayeb, Yazan Alshoibi